

PUMP MAINTENANCE & INSPECTION REPORT

Synthetic test document for petrochemical maintenance-analysis software

Document ID	PM-CRUDE-2026-041	Revision	Rev 2
Plant	North Processing Unit	Area	Crude Transfer
Equipment Tag	P-204A	Equipment Type	Centrifugal Pump
Service	Crude Oil Transfer	Report Date	03-Sep-2026
Maintenance Type	Preventive + Corrective	Priority	High
Operating Hours	18,742 h	Last Overhaul	12-Feb-2025

1. Equipment Description

P-204A is an API 610-style horizontal centrifugal pump used to transfer stabilized crude oil from the intermediate storage section to the desalter feed system. The pump is driven by a 75 kW electric motor through a flexible coupling. A standby pump, P-204B, is installed in parallel.

2. Maintenance Findings

Component	Observation	Severity	Recommended Action
Mechanical seal	Minor leakage observed at seal gland; approximately 3-5 drops/min.	High	Replace seal during next controlled shutdown.
Bearing DE	Temperature reached 78°C; normal operating range is 55-70°C.	Medium	Inspect bearing lubrication and alignment.
Bearing NDE	Temperature 64°C; no abnormal noise.	Low	Continue monitoring.
Coupling	Slight angular misalignment measured at 0.09 mm/100 mm.	Medium	Perform laser alignment.
Pump casing	No visible cracks or deformation; paint deterioration around drain point.	Low	Clean, inspect and repaint.
Baseplate	Four anchor bolts secure; one bolt shows corrosion.	Medium	Clean and replace corroded hardware if required.

3. Operating Condition Data

Parameter	Normal Range	Observed	Status
Discharge pressure	8.0–9.5 bar	8.2 bar	Normal
Suction pressure	1.0–2.0 bar	1.3 bar	Normal
Flow rate	145–175 m³/h	151 m³/h	Normal
Motor current	90–115 A	109 A	Monitor
DE bearing temperature	55–70°C	78°C	Abnormal
Vibration velocity	< 4.5 mm/s	5.8 mm/s	Abnormal
Seal leakage	< 1 drop/min	3–5 drops/min	Abnormal

4. Vibration Analysis

Vibration measurements indicate an increase at the drive-end bearing. Overall vibration was 5.8 mm/s RMS, above the site alert level of 4.5 mm/s RMS. The spectrum showed a dominant component near 1× running speed with a smaller 2× component. The pattern is consistent with possible coupling misalignment, although bearing condition and foundation stiffness should also be checked.

Measurement Point	Horizontal	Vertical	Axial	Overall RMS
Motor DE	3.9 mm/s	3.1 mm/s	2.7 mm/s	5.1 mm/s
Pump DE	4.5 mm/s	3.8 mm/s	3.0 mm/s	5.8 mm/s
Pump NDE	2.7 mm/s	2.4 mm/s	2.1 mm/s	3.6 mm/s

5. Maintenance Work Performed

- Isolated P-204A and verified zero-energy condition according to site lockout/tagout procedure.
- Cleaned the seal gland area and confirmed leakage source.
- Checked coupling guard and visually inspected flexible coupling element.
- Checked bearing lubrication level and replenished approved lubricant by 120 mL.
- Measured shaft/coupling alignment and recorded angular offset of 0.09 mm/100 mm.
- Inspected baseplate and anchor bolts; one corroded anchor bolt was flagged for replacement.

6. Risk Assessment

Risk	Likelihood	Consequence	Risk Level	Control
Seal failure / hydrocarbon leakage	Medium	High	HIGH	Replace mechanical seal; inspect frequently.
Bearing overheating	Medium	Medium	MEDIUM	Check lubrication and temperature trend.
Excessive vibration	Medium	Medium	MEDIUM	Correct alignment and recheck vibration.
Loss of containment	Low	High	MEDIUM	Maintain leak detection and housekeeping.

7. Recommended Actions

1. Schedule mechanical-seal replacement at the earliest controlled shutdown.
2. Perform precision laser alignment of motor and pump shafts after seal work.
3. Inspect DE and NDE bearings for wear and verify lubricant condition.
4. Repeat vibration measurements after alignment; target overall vibration below 4.5 mm/s RMS.
5. Trend DE bearing temperature daily until it returns to the normal range.
6. Inspect the corroded anchor bolt and replace it if cross-section loss is confirmed.
7. Review P-204B standby readiness before taking P-204A out of service.

8. Overall Assessment

Condition: ATTENTION REQUIRED. P-204A remains operational, but elevated vibration, drive-end bearing temperature, and mechanical-seal leakage indicate developing maintenance risk. Continued operation should be supported by increased monitoring and a planned intervention. This synthetic report intentionally contains normal, abnormal, maintenance, safety, and risk-related information to exercise document-analysis, extraction, classification, and recommendation features.

Prepared by: Reliability & Maintenance Engineering | Reviewed by: Maintenance Supervisor

Note: All equipment names, measurements, limits, and observations in this document are fictional test data.